Response to Reviewer 9v2E

Structure-Aware Prediction of PROTAC-Mediated Protein Degradability via Graph Neural Networks

ACM-BCB 2026 Submission

Rating: 4 — Borderline Accept

We thank Reviewer 9v2E for the constructive feedback. We have substantially revised the manuscript; all changes are highlighted in blue in the revised paper. Line numbers below refer to the revised `paper.tex`.

C1: The dataset is relatively small (3,101 samples, 155 unique proteins, 10 E3 ligases). The target-unseen effective diversity is much closer to the number of unique proteins, contributing to seed sensitivity.

Response: We fully agree. We have added a prominent "Dataset characteristics and limitations" paragraph *before* any results are presented:

> *"PROTAC-8K contains 3,101 labeled samples spanning only 155 unique protein targets... The E3 ligase distribution is highly imbalanced: CRBN accounts for 62% of samples, VHL for 35%, and the remaining eight E3s combined represent only 3%. These constraints fundamentally bound generalization performance..."*

We also note that PROTAC-8K is the only publicly available benchmark of this scale, and we have no external validation dataset—both stated as explicit limitations.

[Paper reference: Lines 431–436 (new "Dataset characteristics and limitations" paragraph, §4 Results, before any performance numbers); Limitation item 1 at line 766 (small dataset); Limitation item 2 at line 767 (E3 imbalance); Limitation item 6 at line 771 (no external validation).]

C2: Structural validation is limited. Predictions are not supported by docking, ternary-complex modeling, MD, or free energy calculations.

Response: We agree this is a limitation. It is now discussed explicitly in the Limitations paragraph:

> *"Predictions rely on static AlphaFold structures without physics-based validation (docking, MD, free energy calculations). The BRD4 case study (K374) provides one anecdotal positive control."*

We note an inherent tension: ternary-complex modeling requires knowing the PROTAC linker geometry, which is unavailable in our pre-synthesis setting. Post-hoc structural validation against experimentally determined ternary complexes (where available) is an important future direction.

[Paper reference: Limitation item 5 at line 770 ("No structural validation"); BRD4 case study (§4.5, lines 686–703) provides partial structural validation via lysine attention weight analysis on a known target.]

C3: No compelling biological case study showing why a specific protein is predicted degradable, which lysines drive the prediction, or how predictions align with literature.

Response: We have added a new subsection "Case Study: BRD4 Degradability" (§4.5) that provides exactly this analysis:

- BRD4 (O60885) is predicted highly degradable with CRBN (score: 0.87). [Paper reference: Line 690.]
- Lysine attention weights identify **K374** (attention: 0.12, SASA: 94 Å^2) as the top residue—located in a disordered, solvent-exposed linker between bromodomains. [Paper reference: Lines 691–697.]
- **Literature validation**: mass spectrometry studies by Zengerle et al. (2015) confirm K374 as a major ubiquitination site following JQ1-based PROTAC treatment. [Paper reference: Lines 699–702.]
- We acknowledge this is a single successful case and that systematic validation across more targets would strengthen confidence. [Paper reference: Line 703.]

[Paper reference: New §4.5 "Case Study: BRD4 Degradability" at lines 686–703.]

Additional: The key target-unseen result is unstable; the paper emphasizes the best seed over the multi-seed result; some biological and E3 generalization claims are only partly supported.

Response: All three points are now addressed:

1. **Instability addressed with 6-seed validation**: We trained 3 additional seeds (789, 1011, 1213), yielding a 6-seed mean of 0.603±0.097—marginally below the gradient boosting baseline (0.607, bootstrap $p = 0.556$). We explicitly recommend ensembling across ≥6 seeds for deployment. [Paper reference: Lines 619–625 ("Six-seed extended validation" paragraph); Figure 4 at line 632; Table 1 at lines 441–458 (both 3-seed and 6-seed rows).]
2. **Best seed de-emphasized**: The multi-seed average is now the primary result throughout; best seed (0.7449) is consistently marked as secondary and parenthetical. [Paper reference: Abstract line 65 ("0.646±0.124 AUROC…(best seed: 0.7449)"); Introduction line 135 ("3-seed mean of 0.646…and 6-seed mean of 0.603"); Table 1 line 444 ("Best Seed" as separate column); Results line 460–462 ("multi-seed averages as primary results…best-seed performance should not be interpreted as typical expectation"); Conclusion line 783 (leads with average).]
3. **E3 claims narrowed**: Changed throughout from "E3 generalization" to "CRBN→VHL transfer." VHL failure (0.396 AUROC) and root cause analysis now prominently discussed. Rare E3 generalization explicitly stated as unsupported. [Paper reference: Abstract line 65 ("CRBN→VHL E3-unseen transfer"); Section heading line 635 ("E3 generalization is limited to major ligases"); Root cause analysis lines 640–648; Per-E3 Table 5 at line 654; Per-E3 Figure 5 at line 668; Conclusion line 783 ("CRBN→VHL transfer (0.811 AUROC)").]

Summary of Changes Relevant to Reviewer 9v2E

Concern	Revision (lines in `paper.tex`)
Small dataset (C1)	§4 L431–436 (dataset limitations paragraph); Limitations L766, L767, L771
No structural validation (C2)	Limitation 5 L770; §4.5 L686–703 (BRD4 partial validation)
No case study (C3)	New §4.5 L686–703 (BRD4 with K374 literature validation)
Seed instability	§4.3 L619–625 (6-seed validation); Table 1 L441–458; Figure 4 L632
Best seed emphasis	Abstract L65; Introduction L135; Table 1 L444; §4.2 L460–462; Conclusion L783
E3 claims too broad	Abstract L65; §4.3 L635–648; Table 5 L654; Figure 5 L668; Conclusion L783